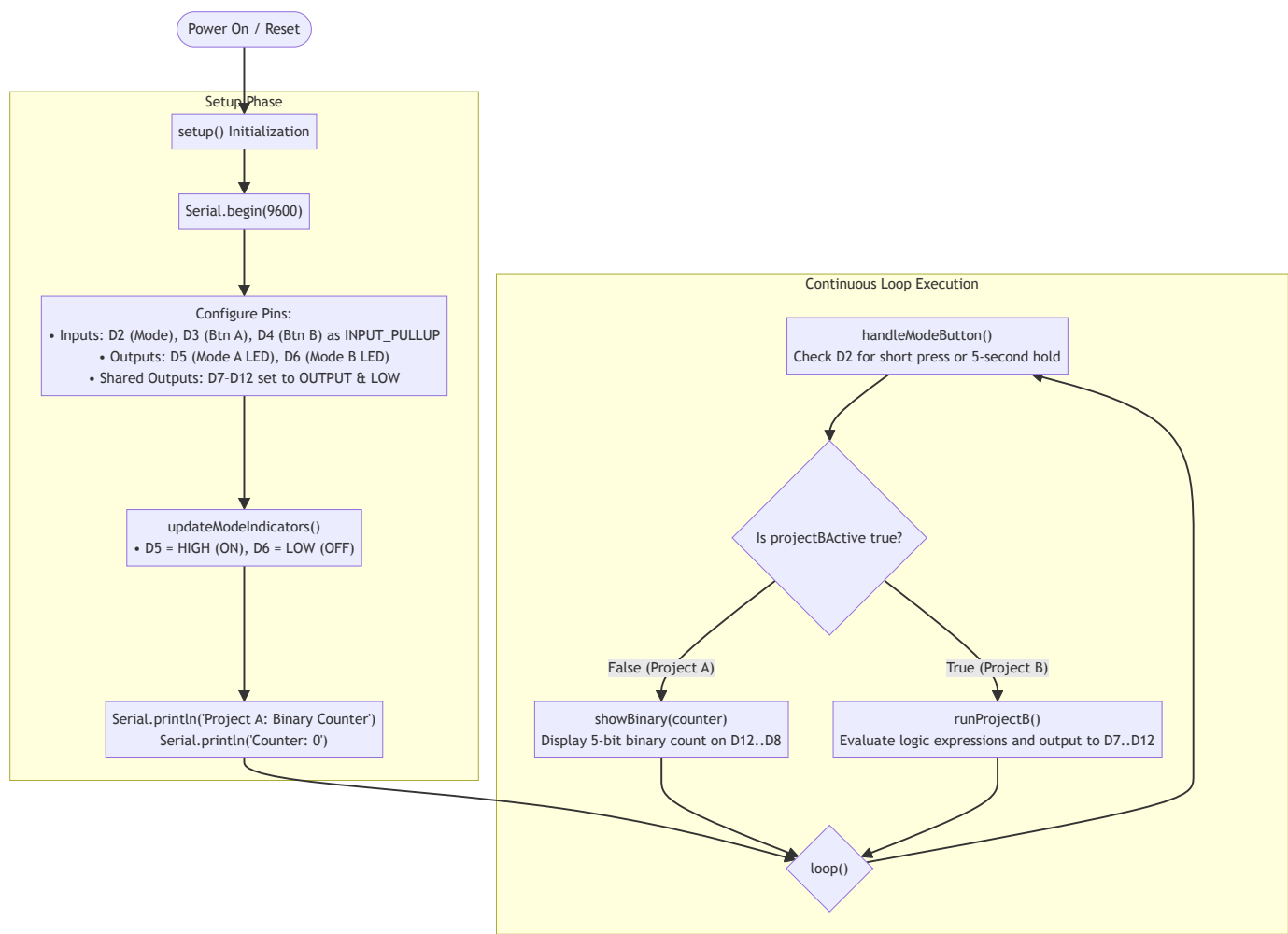


Project 1 Flowcharts

Arduino Program Architecture & Logic Breakdown (MAKR 100)

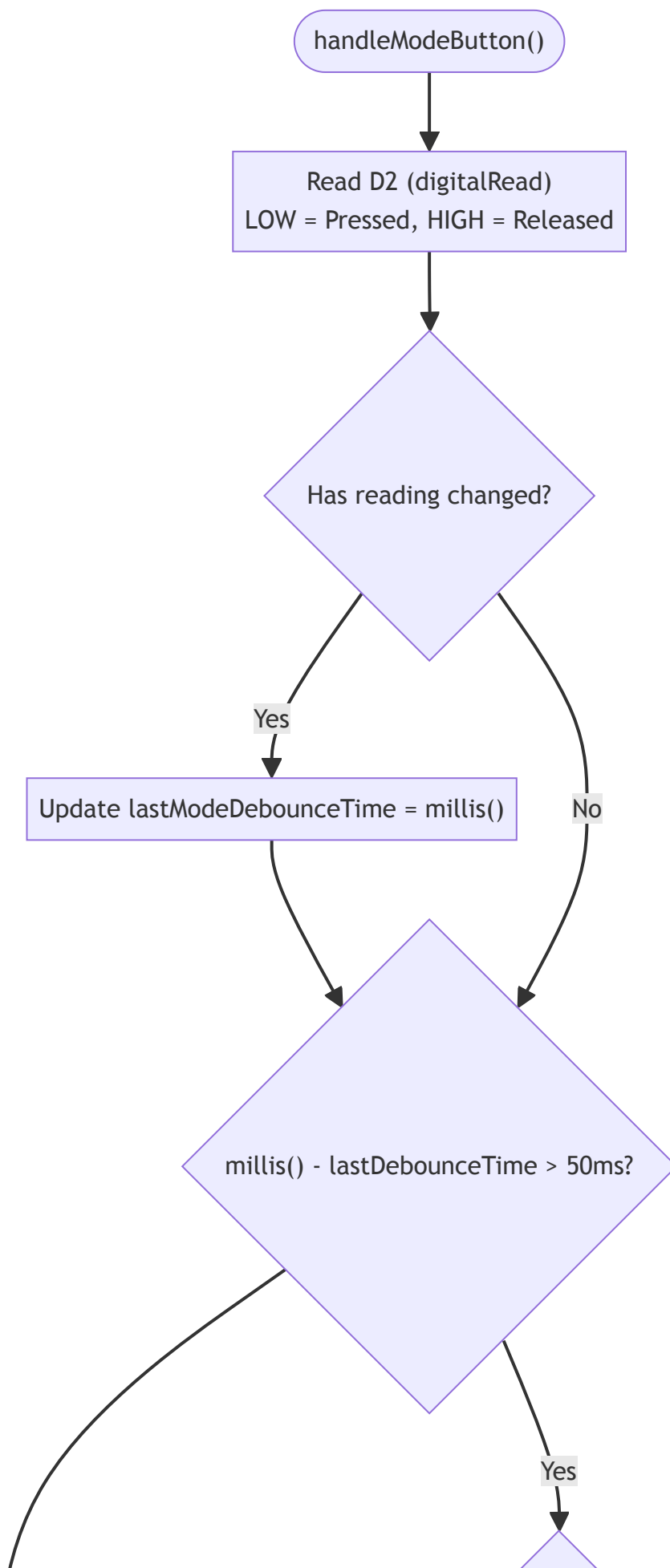
1. Master System Loop

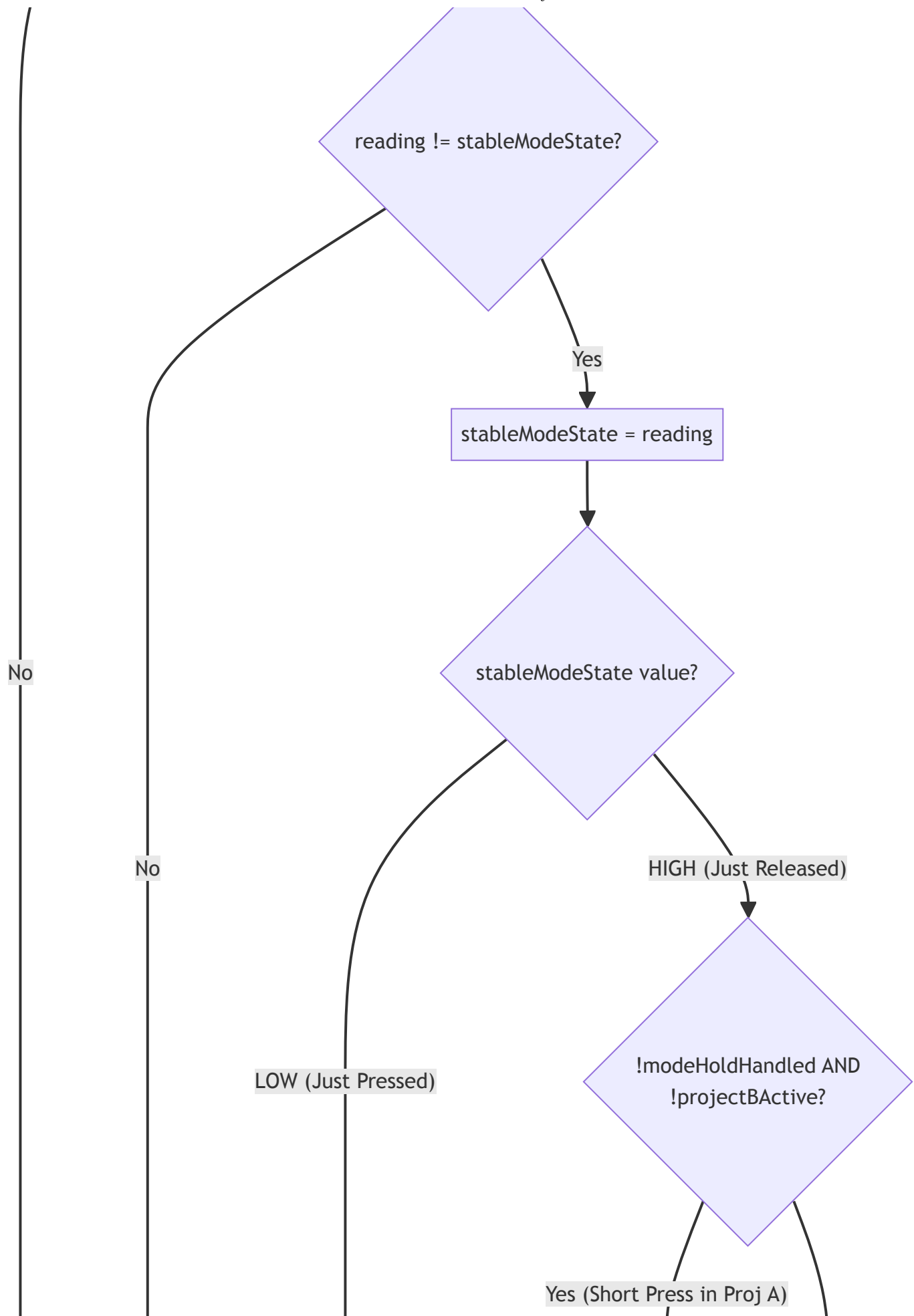
Full lifecycle from setup() to loop() state dispatching



2. Mode Button & Hold/Press Handler

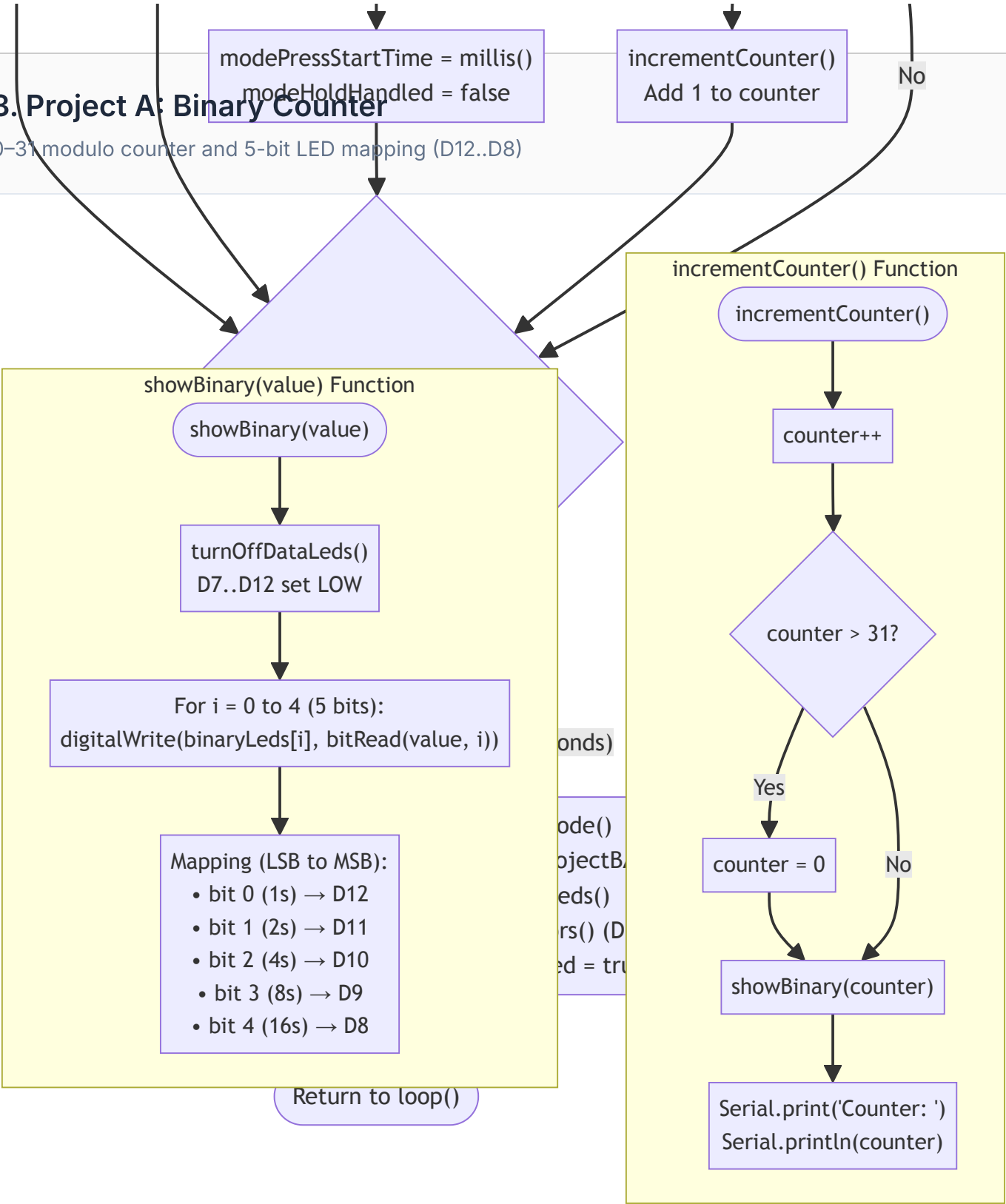
50ms software debounce, 5-second hold switch, and short-press counter increment





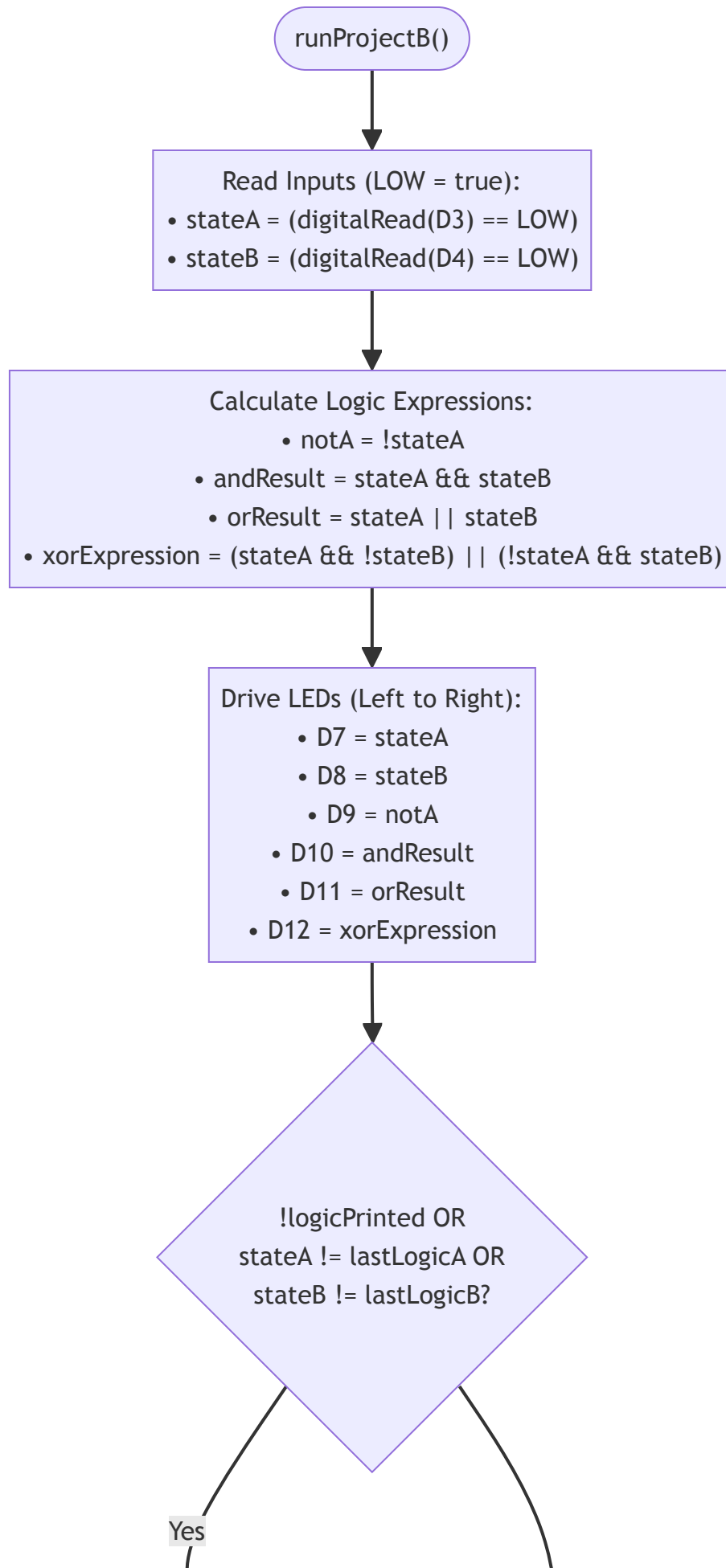
3. Project A: Binary Counter

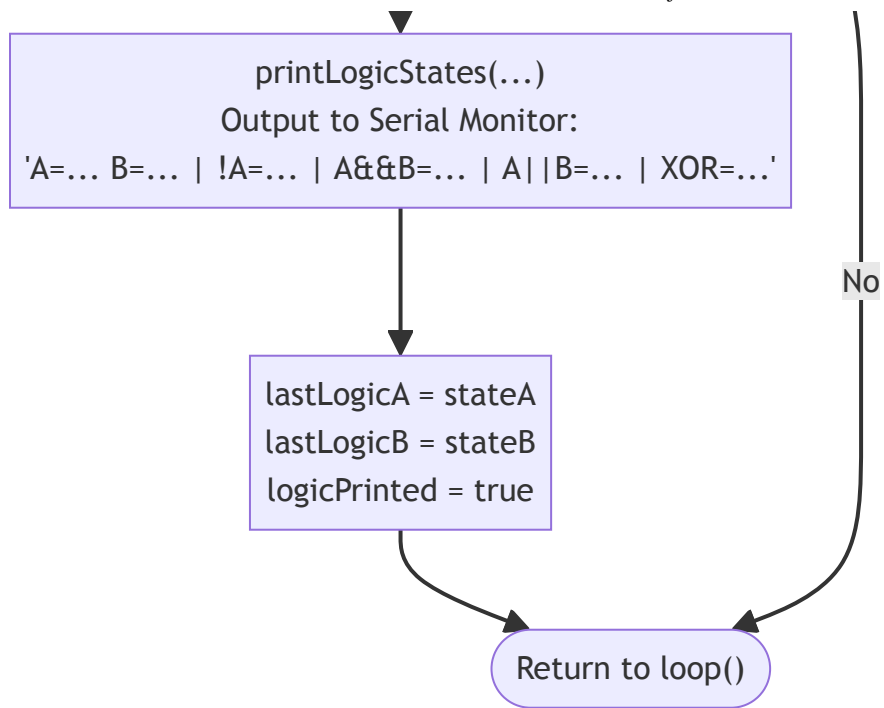
0-31 modulo counter and 5-bit LED mapping (D12..D8)



4. Project B: Logic Gate Operators

Read inputs A & B, evaluate Boolean gates, and drive outputs D7..D12





Hardware Pin Reference Map

Arduino Uno connection specifications for both operating modes

Pin	Project A (Binary Counter)	Project B (Logic Gates)	Mode / Electrical
D2	Short press to increment counter	Hold 5s to return to Project A	INPUT_PULLUP (Pressed = LOW)
D3	(Unused)	Button A	INPUT_PULLUP (Pressed = LOW)
D4	(Unused)	Button B	INPUT_PULLUP (Pressed = LOW)
D5	Active Mode LED (ON)	OFF	OUTPUT
D6	OFF	Active Mode LED (ON)	OUTPUT
D7	OFF	Indicator A	OUTPUT
D8	Bit 4 (16s bit)	Indicator B	OUTPUT
D9	Bit 3 (8s bit)	NOT A (!A)	OUTPUT
D10	Bit 2 (4s bit)	AND (A && B)	OUTPUT
D11	Bit 1 (2s bit)	OR (A B)	OUTPUT
D12	Bit 0 (1s bit / LSB)	XOR ((A && !B) (!A && B))	OUTPUT